

Boundary-Aware Training for Dental Radiograph Instance Segmentation

Technical report

Abstract

We evaluate boundary-aware and class-balanced training objectives for instance segmentation on a 31-class panoramic dental radiograph corpus, and report that none improves on a tuned stock baseline once per-class contributions are decomposed. The result is not a null for want of trying: seven ablation cells at matched budget, four published comparator losses, a second backbone, an architectural change to the mask head, and a downstream clinical endpoint all land within seed-level spread. Three things make the null informative. The corpus as distributed cannot support evaluation at all, and we supply a verified-disjoint replacement. We measure the ceiling the mask representation imposes, independent of any model. And we show that ceiling is *not* the binding constraint, which correctly predicts the failure of a higher-resolution mask head.

1 The Corpus Cannot Be Evaluated as Distributed

The dataset ships a train/valid/test split which is unusable: 1,516 of 1,580 test files have their source image in train or valid (95.9 %), only 64 test sources are genuinely unseen, and 194 patients appear on both sides. Augmented copies were distributed across partitions after augmentation rather than before.

Two further defects: a spurious `croen` category at id 12 in the test COCO file shifts every higher class id out of alignment, and `data.yaml` points `test:` at the validation images.

We rebuild patient-grouped and class-stratified (9752 / 2090 / 2090; test coverage 13 $\rightarrow$ 29 of 31 classes) and verify disjointness twice by independent implementations: 0 exact duplicates, 0 NCC-confirmed near-duplicates across 792,811 candidate pairs, 0 shared patients. Maximum NCC is 0.9334 against a 0.98 threshold.

Perceptual hashing alone is not a leakage test on this modality. Every panoramic radiograph is the same anatomy in the same framing, so unrelated studies routinely fall within 5 bits on an 8×8 dHash. Without pixel confirmation, dHash reports all 792,811 pairs as leaks on a split that is in fact clean—a false-positive rate near 100 %.

Statistical power. Sixteen of 31 classes appear in fewer than 10 validation images; two appear in none; eleven tail classes score exactly zero. No long-tail claim is supportable and none is made.

2 The Objective

Let $M : \Omega \rightarrow \{0, 1\}$ be a ground-truth instance mask, $\hat{M}$ the predicted logits, $p = \sigma(\hat{M})$. For odd band width k with $S_k(i)$ the $k \times k$ window at pixel i :

$$\delta_k(x)[i] = \max_{j \in S_k(i)} x[j], \quad (1)$$

$$\varepsilon_k(x)[i] = \min_{j \in S_k(i)} x[j], \quad (2)$$

$$B_k(x) = \delta_k(x) - \varepsilon_k(x). \quad (3)$$

B_k is the morphological gradient, computed as one max-pool using $\varepsilon_k(x) = -\delta_k(-x)$. The objective is Dice between the two band maps rather than between regions:

$$\mathcal{L}_{\text{band}} = 1 - \frac{2 \sum_{\Omega} B_k(p) \odot B_k(M) + \epsilon}{\sum_{\Omega} B_k(p) + \sum_{\Omega} B_k(M) + \epsilon}, \quad (4)$$

added to the stock cropped BCE with weight λ . Throughout $k = 3$, $\lambda = 0.5$, never swept.

Mechanism. $\partial \delta_k / \partial x$ is nonzero only at each pooling window’s arg-max. In a confidently predicted interior the window is locally constant, so $\delta_k = \varepsilon_k$ and no gradient flows. The term acts only on pixels at or beside the predicted edge—measured at **48.5 % nonzero gradient versus 100 % for region losses**. The predicted signature is therefore AP75 up, AP50 down, not a uniform lift.

Novelty, stated honestly. The construction is a synthesis of known components: morphological gradient as a differentiable edge extractor, soft Dice [2] as the overlap objective. Its nearest relative, Boundary IoU [1], is an evaluation metric. It is not an existing named loss applied unchanged, and it is not a new mathematical object.

3 Ablation and Test

arm	configuration	mAP	AP50	AP75	head
S0	reference	0.1055	0.2549	0.0661	0.2815
S1c	inverse-sqrt	0.1065	0.2591	0.0659	0.2836
S2	inv-sqrt + boundary	0.1050	0.2467	0.0730	0.2921
S3	inv-sqrt + copy-paste	0.1042	0.2570	0.0635	0.2836
S4	all three	0.1051	0.2513	0.0675	0.2916

Table 1: Validation, 50 epochs, mask metrics. The grid is cumulative, so $S2 - S0$ conflates weighting with the boundary term. The valid isolation is $S2 - S1c$: AP75 +0.70 pp (+10.7 %), head +0.85 pp (+3.0 %), AP50 −1.23 pp.

Class weighting does nothing, mechanically. YOLO’s task-aligned assigner selects positives by $\text{cls}^\alpha \times \text{IoU}^\beta$. For a class the model never learned, $\text{cls} \approx 0$, so the ground truth receives few positive assignments and re-weighting scales a loss over assignments never made. The bottleneck is upstream of the loss.

Copy-paste hurts, and the mechanism generalises. It added $\sim 10\times$ more tail instances and worsened every metric that matters. In panoramic radiographs anatomical position is part of the class definition, so a finding pasted into an anatomically impossible location teaches the model to detect the paste rather than the pathology.

model (100 epochs)	mAP	AP50	AP75	tail
baseline	0.1051	0.2590	0.0687	0.0365
boundary alone	0.1071	0.2585	0.0680	0.0441
weighting + boundary	0.1007	0.2430	0.0650	0.0261
weighting alone	0.0973	0.2369	0.0612	0.0252

Table 2: Held-out test. The nominal +0.21 pp of the boundary arm decomposes to three classes with 4, 8 and 8 test instances. AP50, AP75 and head are flat.

Validation ordering did not transfer, three times. S1c led on validation and came last on test. The +10.7 % AP75 gain at 50 epochs on validation is −0.37 pp at 100 epochs on test.

4 The Representation Ceiling Is Not the Constraint

Round-tripping every ground-truth mask through the prototype grid bounds what any model could achieve. At the stock 160×160 grid, mean Dice is 0.8963 and **17.7 % of instances cannot reach IoU 0.75**; root canal treatment is capped at 0.622 and caries at 0.721, both below threshold.

But the model achieves Dice **0.6969**—only 78 % of the ceiling it already has. Doubling the grid raises a limit nothing is pressing against.

We tested it: a prototype head fed from P2 (stride 4) rather than P3 (stride 8), using sub-pixel convolution [3], with **878,432 fewer parameters**. Against a matched control at identical budget: −0.72 pp mAP, −0.49 pp AP75. The prediction was wrong and the experiment says so.

5 Metric Failure: Distances on Per-Model Subsets

HD95 and ASSD are undefined on empty masks, so they average over a denominator that differs per model. Averaged per model, the boundary objective appeared **17.3 % better on HD95 and 24.6 % better on ASSD with disjoint intervals**. Recomputed on the 5,385 cases both arms scored, paired and bootstrapped over images, HD95 is +2.30 *worse* with the interval excluding zero, and every other metric is inseparable from zero. Correcting the denominator inverted the sign.

6 Clinical Endpoint

Endpoint: bone-loss area as a percentage of image area. Distance-based and normalized bone-level endpoints are *not computable*—both require cemento-enamel-junction and root-apex landmarks, and the annotations provide a region polygon only; the images carry no pixel-spacing metadata.

Paired per-image absolute error: +0.0154 pp, 95 % CI [−0.0738, +0.1060]. No detectable change. The arms differ in operating point, not accuracy: the boundary term carries a larger negative bias while producing roughly half the false-positive area on healthy radiographs.

7 The Mask Deficit Is Coefficient Prediction, Not Representation

The 20-point gap between what the model achieves (Dice 0.6969) and what its prototype grid allows (0.8963) had no attribution. Prototype-based heads reconstruct each mask as $\sigma(\mathbf{c}^\top P)$, so either the learned 32-dimensional basis cannot represent these shapes, or the basis is adequate and the coefficient head fails to locate the right point in it. Those call for opposite fixes.

We separate them without training. For every ground-truth instance we solve, in closed form, for the coefficient vector that best reconstructs it from the model’s own prototypes,

$$\mathbf{c}^* = \arg \min_{\mathbf{c}} \|P^\top \mathbf{c} - y\|^2, \quad (5)$$

over the instance’s box crop, where $y \in \{-1, +1\}$ is the ground-truth mask. Thresholding $P^\top \mathbf{c}^*$ gives the best mask *any* coefficient head could produce from that basis.

quantity	Dice
grid ceiling (what the resolution allows)	0.8963
oracle coefficients, full resolution	0.8659
oracle coefficients, at prototype resolution	0.9636
what the trained model achieves	0.6969

source of the 0.1994 gap	Dice lost	share
prototype basis cannot represent it	0.0304	15 %
coefficient head fails to find it	0.1690	85 %

Table 3: 4123 instances. The basis is not the constraint: handed optimal coefficients the model jumps from 0.70 to 0.87.

This closes the loop on every negative result above. Boundary terms and comparator losses reshape the *objective*; higher prototype resolution enlarges the *representation*. Neither is the binding constraint, which is why neither moved the number, and why the P2-fed head performed worse rather than better—it added resolution to a basis that was never the limit.

The head capacities make this concrete. In YOLOv8x-seg the coefficient branch bottlenecks 320 to 80 channels and carries 1.33 M parameters, against 2.26 M for the prototype generator and 7.41 M for classification. The smallest of the three heads is responsible for 85 % of the mask deficit.

8 Noise Floor (Measured)

The reference configuration was trained at three seeds, everything else identical. Determinism is exact on this pipeline, so seed choice is provably the only source of variation.

seed	mAP	AP50	AP75	head	tail
42	0.1055	0.2549	0.0661	0.2815	0.0101
1337	0.1056	0.2573	0.0679	0.2837	0.0117
2024	0.1037	0.2568	0.0629	0.2823	0.0105
sd	0.0010	0.0013	0.0025	0.0011	0.0008

Table 4: Noise floor at 2 sd: mAP ± 0.21 , AP50 ± 0.26 , AP75 ± 0.50 , head ± 0.22 , tail ± 0.17 pp.

Three consequences, and they are not the same verdict. The seven-arm ablation spans a 0.36 pp range, *inside* the band—selecting a winner on mAP was selecting noise. The best segmentation arm on test is +0.21 pp, sitting *exactly* at the threshold. But the isolated boundary effect on AP75 is +0.70 pp against a ± 0.50 pp floor, so it **exceeds** noise on validation at 50 epochs, and then reverses on test at 100 epochs. That is a **transfer failure, not a noise result**: the objective does what its mechanism predicts under the conditions it was measured in, and that does not survive a change of split and schedule.

9 Conclusion and Limitations

On this corpus, boundary-aware and class-balanced objectives do not improve instance segmentation beyond seed-level variation. What the work establishes instead is that the distributed split cannot support evaluation; that perceptual hashing alone is not a leakage test here; that the representation ceiling is real but not binding; that distance metrics on per-model subsets invert their sign; and that anatomical position is part of the class definition in radiographs.

Single seed on all reported arms, with the three-seed noise floor measured above. Single corpus, single backbone family. No prior published results exist for this dataset. No long-tail claim, because the split cannot support one.

References

- [1] Bowen Cheng, Ross Girshick, Piotr Dollár, Alexander C. Berg, and Alexander Kirillov. Boundary IoU: Improving object-centric image segmentation evaluation. In *CVPR*, 2021.
- [2] Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi. V-Net: Fully convolutional neural networks for volumetric medical image segmentation. In *3DV*, 2016.
- [3] Wenzhe Shi, Jose Caballero, Ferenc Huszár, Johannes Totz, Andrew P. Aitken, Rob Bishop, Daniel Rueckert, and Zehan Wang. Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network. In *CVPR*, 2016.